5-G-Product of Array elements-Minimum

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer:(penalty regime: 0 %)

Feedback

#include <stdio.h>

#include <stdlib.h>

```
int ascending(const void *a, const void *b)
{
   return (*(int *)a - *(int *)b);
}

int descending(const void *a, const void *b)
{
   return (*(int *)b - *(int *)a);
}

int main()
{
   int n;
   scanf("%d", &n);

   int A[n], B[n];

   for (int i = 0; i < n; i++)
      scanf("%d", &A[i]);

   for (int i = 0; i < n; i++)
      scanf("%d", &B[i]);

   qsort(A, n, sizeof(int), ascending);
   qsort(B, n, sizeof(int), descending);

   long long sum = 0;
```

```
    for (int i = 0; i < n; i++)
    {
        sum += (long long)A[i] * B[i];
    }

    printf("%lld", sum);

    return 0;
}
```

	Input	Expected	Got	
	3 1 2 3 4 5 6	28	28	
	4 7 5 1 2 1 3 4	22	22	

	Input	Expected	Got	
	1			
	5 20 10 30 10 40 8 9 4 3 10	590	590	

Passed all tests!

Divide and Conquer